Oracle and grouping properties of the Liu-SCAD estimator with applications to low and high-dimensional regression

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Abstract

Classical ℓ_2-regularization approaches, such as ridge regression and the Liu estimator, improve prediction accuracy in the presence of multicollinearity but do not perform variable selection. To address this limitation, we introduce *Liu-SCAD*, a new penalized regression estimator that integrates the bias-reduction property of the Liu estimator with the sparsity-inducing Smoothly Clipped Absolute Deviation (SCAD) penalty. The proposed method achieves a favourable compromise between shrinkage, bias control, and variable selection. We establish its theoretical properties, including oracle consistency and the grouping effect, under high-dimensional settings. Extensive simulation studies and applications to real datasets indicate that Liu-SCAD consistently outperforms or rivals existing methods such as Lasso, SCAD and Liu-Lasso particularly in multicollinear and high-dimensional scenarios. These results highlight Liu-SCAD as a competitive and robust tool for modern regression analysis.

Keywords: Regularization; Variable selection; High-dimensional data; Liu estimator; Penalized regression.

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1. Introduction

Regression analysis remains a foundational tool in statistics, extensively studied for its versatility and widespread applicability across numerous scientific disciplines [16]. Its primary goal is to understand how a response variable depends on one or more explanatory variables. The linear regression model (LRM), which assumes a linear relationship between the response and predictors, is typically estimated using the ordinary least squares (OLS) method. While OLS provides optimal estimates under classical assumptions, it suffers notable limitations in the presence of multicollinearity or when the number of predictors exceeds the sample size. In such cases, the design matrix becomes singular or nearly singular, leading to unstable estimates with inflated variance. Moreover, OLS fails to produce unique solutions in high-dimensional settings, which are increasingly common in modern data-driven fields such as genomics, neuroimaging, and finance [1, 7, 21].

High-dimensional data analysis often begins with a large number of candidate predictors to mitigate model bias. However, this introduces challenges related to overfitting and interpretability [5]. Traditional variable selection techniques, such as stepwise procedures and best subset selection, are commonly employed to address this. Despite their practical utility, these methods often ignore the stochastic nature of model selection, leading to theoretical and computational instability. Notably, best subset selection lacks robustness, as highlighted in Breiman [2], due to its sensitivity to small changes in the data.

To address these shortcomings, penalized least squares (PLS) methods have gained prominence. By incorporating a penalty term into the loss function, PLS methods simultaneously achieve regularization and estimation. Convex penalties, such as the ℓ_2 norm used in ridge regression and its modification in the Liu estimator, stabilize coefficient estimates through shrinkage but do not yield sparsity [11, 14]. In contrast, nondifferentiable or nonconvex penalties like the ℓ_1 norm used in the lasso [18] enable variable selection by shrinking some coefficients exactly to zero. This sparsity-promoting property has led to the development of a broad class of penalized methods, including the bridge penalty [6], Elastic Net (which combines ridge and lasso penalties) [26], and smoothly clipped absolute deviation (SCAD) [5].

The SCAD penalty was designed to overcome the limitations of the lasso by reducing bias in large coefficients while achieving consistent variable selection. SCAD thus provides a balance between sparsity, continuity, and

estimation efficiency, making it well-suited for high-dimensional problems [12]. On the other hand, the Liu estimator was introduced as a bias-reduced alternative to ridge regression, particularly effective in the presence of multicollinearity [17]. Although SCAD and Liu estimators address different aspects of regularization, they do not simultaneously resolve multicollinearity and consistent variable selection.

Hybrid approaches have therefore emerged. Elastic Net combines ridge and lasso penalties but does not satisfy the oracle property and may excessively shrink large coefficients. More recently, Liu-Lasso has integrated Liu-type shrinkage with the lasso penalty; however, it still inherits the drawbacks of lasso, such as bias and the lack of oracle property in correlated settings.

In this study, we propose a new estimator, called Liu-SCAD, that combines the bias-reduction ability of the Liu estimator with the sparsity-inducing SCAD penalty. Unlike Elastic Net and Liu-Lasso, the Liu-SCAD estimator simultaneously addresses multicollinearity and consistent variable selection. We establish its oracle and grouping properties under general non-orthogonal designs, showing that it can correctly identify the true model and estimate nonzero coefficients efficiently. In addition, we develop a coordinate descent algorithm specifically tailored to solve the Liu-SCAD optimization problem, making the method computationally feasible for large datasets. Extensive simulation studies and real-data applications further demonstrate that Liu-SCAD often outperforms or matches classical methods such as Lasso, SCAD and Liu-Lasso, particularly in high-dimensional and highly correlated environments.

In summary, this study advances the development of penalized regression techniques for high-dimensional linear models by proposing the Liu-SCAD estimator. Section 2 provides a brief overview of existing variable selection methods and introduces the formulation of the Liu-SCAD approach. In Section 3, we establish the asymptotic properties of the proposed estimator under general, non-orthogonal designs, including the oracle and grouping properties, under suitable regularity conditions. Section 4 presents results from extensive simulation studies that assess the finite-sample performance of Liu-SCAD in comparison with competing methods. Section 5 demonstrates the practical utility of the method through applications to real-world datasets. Concluding remarks are provided in Section 6, and detailed technical proofs are deferred to the Appendix.

2. Model and Penalized least squares

The general LRM in matrix form is given by:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{1}$$

where $\mathbf{y} \in \mathbb{R}^n$ is the response vector, $\mathbf{X} \in \mathbb{R}^{n\times p}$ is the predictor design matrix, $\boldsymbol{\beta} \in \mathbb{R}^p$ is the vector of the true regression coefficient, and $\boldsymbol{\varepsilon} \in \mathbb{R}^n$ is the vector of error terms with mean zero and variance $\sigma^2\mathbf{I}_n$. The design matrix $\mathbf{X}$ is standardized so that $\sum_{i=1}^n x_{ij}/n = 0$, $\sum_{i=1}^n x_{ij}^2/n = 1$ for $j = 1, 2, \ldots, p$ and the response vector $\mathbf{y}$ is centered so that $\sum_{i=1}^n y_i/n = 0$. This allowed us to fit a regression model without the intercept term. OLS is the solution to the following optimization problem.

$$\hat{\boldsymbol{\beta}}^{LS} = \arg\min_{\boldsymbol{\beta}\in\mathbb{R}^p}\{\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2\} = (X^TX)^{-1}X^Ty. \tag{2}$$

OLS is known to be unbiased and efficient under the classical linear model assumptions, particularly in the absence of multicollinearity. However, when predictors exhibit strong linear dependence, the variance of the OLS can become excessively large, resulting in unstable and potentially misleading coefficient estimates [15]. This challenge is further exacerbated in high-dimensional settings, where the number of predictors exceeds the number of observations, as the design matrix $\mathbf{Z}$ lacks full rank, rendering the least squares solution non-unique [8].

To overcome these limitations, penalized regression methods have emerged as powerful alternatives. By augmenting the OLS criterion with a penalty term, these methods shrink the magnitude of the regression coefficients, thereby reducing variance and improving numerical stability. A general framework for penalized least squares estimation takes the form:

$$\hat{\boldsymbol{\beta}}(\lambda, \nu) = \arg\min_{\boldsymbol{\beta}\in\mathbb{R}^p}\left\{\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda\, p_\nu(\boldsymbol{\beta})\right\}, \tag{3}$$

where $p_\nu(\cdot)$ denotes a penalty function (e.g., ℓ_1, ℓ_2) and $\lambda > 0$ governs the strength of regularization. This formulation encompasses a broad class of methods designed to strike a balance between prediction accuracy, model complexity, and interpretability.

Ridge regression, introduced by Hoerl and Kennard [11], is one of the earliest and most widely adopted penalized estimation techniques. Ridge

regression addresses multicollinearity by adding an ℓ_2-norm penalty to the loss function, yielding the estimator:

$$\hat{\boldsymbol{\beta}}^{k} = \arg\min_{\boldsymbol{\theta}} \left\{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \sum_{j=1}^{p_n} \beta_j^2 \right\} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}, \tag{4}$$

where $\lambda \geq 0$ is a tuning parameter that controls the degree of regularization. Although ridge regression improves estimation in the presence of multicollinearity and ensures a unique solution even in high-dimensional settings, it does not induce sparsity; all coefficients are shrunk toward zero, but none are set exactly to zero. This lack of variable selection can hinder model interpretability in settings where identifying a parsimonious subset of predictors is of interest.

As an alternative within the class of biased estimators, the Liu estimator (LE) [14] offers a compelling compromise between variance reduction and interpretability. LE introduces a tuning parameter $0 \leq d < 1$ that blends OLS and Stein-type shrinkage estimators, leading to improved performance when the predictors are highly correlated. Formally, LE augments the ridge formulation by incorporating a scaled version of the OLS, yielding the closed-form solution:

$$\hat{\boldsymbol{\beta}}^{(\mathrm{liu})}(d) = \arg\min_{\boldsymbol{\beta} \in \mathbb{R}^p} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \frac{1}{2} \left\| \boldsymbol{\beta} - d\hat{\boldsymbol{\beta}}_0^{\mathrm{ls}} \right\|_2^2 \right\} = \left(\mathbf{X}^\top \mathbf{X} + \mathbf{I}_p \right)^{-1} \left(\mathbf{X}^\top \mathbf{y} + \hat{\boldsymbol{\beta}}^{s} \right), \tag{5}$$

where $\hat{\boldsymbol{\beta}}^{s} = d\,\hat{\boldsymbol{\beta}}^{\mathrm{LS}}$ denotes a Stein-type shrinkage estimator based on the ordinary least squares (OLS) solution. In high-dimensional settings where the OLS estimator is not uniquely defined, we instead set $\hat{\boldsymbol{\beta}}^{s} = d\,\hat{\boldsymbol{\beta}}^{\mathrm{k}}$, using the ridge estimator $\hat{\boldsymbol{\beta}}^{\mathrm{k}}$ as the shrinkage target. This formulation enables the Liu estimator to integrate the stabilization benefits of ridge regression with the adaptability of shrinkage-based approaches, offering a compelling and interpretable alternative for handling multicollinearity. However, similar to ridge regression, the Liu estimator does not yield sparse solutions, as none of the coefficient estimates are driven exactly to zero. This limits its effectiveness in variable selection tasks where model parsimony is desired.

The increasing availability of high-dimensional data, such as gene expression profiles from microarray experiments, has heightened the need for regression techniques that promote sparsity in model estimation [23]. One of

the earliest and most widely used methods in this context is the least absolute shrinkage and selection operator (Lasso), which simultaneously performs variable selection and coefficient estimation by imposing an ℓ_1-norm penalty on the regression coefficients. This penalty encourages sparsity by shrinking some coefficients exactly to zero. The Lasso estimator is formally defined as

$$\hat{\boldsymbol{\beta}}^{\text{lasso}} = \arg\min_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}, \tag{6}$$

where $\lambda \geq 0$ is a regularization parameter that governs the level of shrinkage. Larger values of λ result in sparser solutions, with more coefficients exactly zero, thereby facilitating automatic variable selection. The limitations of the Lasso estimator in handling highly correlated predictors have been well documented. In particular, when a group of covariates exhibits strong multicollinearity, the Lasso tends to arbitrarily select only one variable from the group, disregarding the rest [26]. This behavior arises from the ℓ_1-penalty, which, although effective in inducing sparsity, lacks the grouping property and introduces bias in estimating large coefficients.

To address these shortcomings, researchers have proposed alternative penalty functions that extend beyond the ℓ_1-norm. One notable contribution is the smoothly clipped absolute deviation (SCAD) penalty, introduced by Fan and Li [5] is defined for a scalar coefficient $\theta \in \mathbb{R}$, a regularization parameter $\lambda > 0$, and a shape parameter $a > 2$, as:

$$p_\lambda(\theta; a) = \begin{cases} \lambda|\theta|, & \text{if } |\theta| \leq \lambda, \\ \frac{-\theta^2 + 2a\lambda|\theta| - \lambda^2}{2(a-1)}, & \text{if } \lambda < |\theta| \leq a\lambda, \\ \frac{(a+1)\lambda^2}{2}, & \text{if } |\theta| > a\lambda. \end{cases}$$

The SCAD-penalized least squares estimator is then defined as the solution to the following optimization problem:

$$\hat{\boldsymbol{\beta}}^{\text{SCAD}} = \arg\min_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \sum_{j=1}^{p} p_\lambda\left(|\beta_j|; a\right) \right\}. \tag{7}$$

Here, the term

$$\sum_{j=1}^{p} p_\lambda\left(|\beta_j|; a\right)$$

applies the SCAD penalty elementwise to the components of $\boldsymbol{\beta}$, encouraging sparse solutions while avoiding excessive penalization of large coefficients. The penalty function is non-convex and designed to satisfy the oracle property under suitable regularity conditions. The SCAD penalty was specifically designed to satisfy three desirable properties: (i) it is singular at the origin to promote sparsity, (ii) it is continuous to yield stable estimators, and (iii) it is bounded for large coefficients, thus reducing estimation bias. Notably, SCAD has been shown to satisfy the oracle property, selecting the correct subset of variables with probability tending to one and yielding asymptotically normal estimators for the nonzero coefficients as if the zero coefficients were known in advance. Despite these theoretical strengths, like Lasso, it does not inherently account for multicollinearity.

To better address multicollinearity, Liu [14] proposed a shrinkage estimator that introduces a bias in exchange for reduced variance. Genc [8] extended this idea by integrating the Liu estimator with the Lasso penalty, resulting in the Liu-Lasso estimator. This hybrid approach retains the regularization benefits of Lasso while incorporating the stabilizing effects of the Liu estimator.

The Liu-Lasso estimator is defined as:

$$\hat{\boldsymbol{\beta}}^{\mathrm{ll}}(\lambda, d) = \arg\min_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ \frac{1}{2n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \frac{1}{2}\|\boldsymbol{\beta} - d\hat{\boldsymbol{\beta}}^{\mathrm{ls}}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1 \right\}, \tag{8}$$

where $\hat{\boldsymbol{\beta}}^{\mathrm{ls}}$ is the OLS estimator, λ is a nonnegative tuning parameter, and d controls the degree of shrinkage toward the OLS solution.

While the Liu-Lasso estimator integrates the stabilizing effect of the Liu shrinkage with the sparsity-inducing Lasso penalty, it inherits a notable limitation of the Lasso-*the failure to satisfy the oracle property* [5, 25]. Specifically, the ℓ_1-penalty introduces substantial bias in the estimation of large coefficients and may lead to inconsistent variable selection, particularly in high-dimensional regimes with correlated predictors. As a result, Liu-Lasso may not reliably distinguish between signal and noise when the true underlying model is sparse.

To address this shortcoming, we propose the *Liu-SCAD estimator*, which fuses the Liu shrinkage formulation with the smoothly clipped absolute deviation (SCAD) penalty. The SCAD penalty is designed to satisfy the oracle property under suitable regularity conditions. By applying a non-convex, adaptive penalization, Liu-SCAD reduces estimation bias for large coefficients and improves model selection accuracy, thereby achieving *asymptotic*

consistency in variable selection and unbiased estimation. In this way, Liu-SCAD combines the stability of Liu-type shrinkage with the theoretical optimality of SCAD, offering a more robust alternative to Liu-Lasso in sparse high-dimensional settings. Specifically, we replace the ℓ_1-penalty in the Liu-Lasso formulation with the SCAD penalty, resulting in the following objective function:

$$\hat{\boldsymbol{\beta}}^{\text{Liu-SCAD}}(\lambda, d) = \arg\min_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ \frac{1}{2n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \frac{1}{2}\|\boldsymbol{\beta} - d\hat{\boldsymbol{\beta}}^{\text{ls}}\|_2^2 + \sum_{j=1}^{p} p_\lambda^{\text{SCAD}}(|\beta_j|) \right\}, \tag{9}$$

where $p_\lambda^{\text{SCAD}}(\cdot)$ denotes the SCAD penalty function applied element-wise. It is defined as:

$$p_\lambda^{\text{SCAD}}(\theta) = \begin{cases} \lambda|\theta|, & \text{if } |\theta| \le \lambda, \\ \frac{-\theta^2 + 2a\lambda|\theta| - \lambda^2}{2(a-1)}, & \text{if } \lambda < |\theta| \le a\lambda, \\ \frac{(a+1)\lambda^2}{2}, & \text{if } |\theta| > a\lambda, \end{cases} \tag{10}$$

with $a > 2$, typically set to 3.7.

This hybrid estimator is particularly well-suited for high-dimensional data structures in which both sparsity and multicollinearity are prevalent. By blending SCAD's selection capabilities with Liu's bias-control mechanism, the Liu-SCAD estimator aims to provide a more stable, interpretable, and efficient solution in modern regression settings.

2.1. Coordinate Descent for Liu-SCAD

To compute the Liu-SCAD estimates, we adopted a coordinate descent (CD) algorithm, an efficient strategy for solving penalized regression problems with separable penalties. The key idea is to iteratively minimize the objective with respect to one coordinate at a time while holding the others fixed. This is particularly useful for nondifferentiable penalties such as SCAD, where a global closed-form solution is unavailable.

Given the objective function:

$$\hat{\boldsymbol{\beta}}^{\text{Liu-SCAD}} = \arg\min_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ \frac{1}{2n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \frac{1}{2}\|\boldsymbol{\beta} - d\hat{\boldsymbol{\beta}}^{\text{LS}}\|_2^2 + \sum_{j=1}^{p} p_\lambda^{\text{SCAD}}(|\beta_j|) \right\}, \tag{11}$$

we isolate the update for a single coefficient β_j. Define the partial residual as:

$$r_j = \mathbf{y} - \sum_{k \neq j} X_k \beta_k, \tag{12}$$

and compute the coordinate-wise subproblem:

$$\min_{\beta_j} \left\{ \frac{1}{2n} \|r_j - X_j \beta_j\|_2^2 + \frac{1}{2}(\beta_j - d\hat{\beta}_j^{\mathrm{LS}})^2 + p_\lambda^{\mathrm{SCAD}}(|\beta_j|) \right\}. \tag{13}$$

The closed-form update for β_j is obtained via:

$$\beta_j \leftarrow \frac{\text{SCAD-threshold}(z_j + d\hat{\beta}_j^{\mathrm{LS}}, \lambda)}{\omega_j + 1}, \tag{14}$$

where $z_j = \frac{1}{n} X_j^\top r_j$ and $\omega_j = \frac{1}{n}\|X_j\|^2$. Here, SCAD-threshold denotes the piecewise thresholding operator associated with the SCAD penalty [5].

This process is repeated cyclically on all coordinates until convergence. Warm starts are used to initialize $\boldsymbol{\beta}$ based on solutions from nearby values of λ, accelerating convergence. Convergence is typically determined by monitoring the relative change in coefficients or the objective value.

Coordinate descent is particularly suitable here because the Liu term is quadratic and smooth. At the same time, the SCAD penalty, though nonconvex, is separable and locally smooth outside of a small neighborhood around zero. This structure ensures that each subproblem remains tractable and that the algorithm converges to a stationary point, as supported by empirical studies and theoretical guarantees in Tseng[19].

2.2. Orthonormal Design Case

The behavior of the Liu-SCAD estimator is particularly interpretable when the design matrix satisfies the orthonormal condition $\frac{1}{\sqrt{n}}\mathbf{X}^\top \mathbf{X} = \mathbf{I}_p$. Under this scenario, the individual coordinate updates simplify, and the solution admits a Stein-type closed form. Following the logic of the Liu-Lasso framework, the Liu-SCAD estimator under orthonormality can be expressed as:

$$\hat{\beta}_j(\lambda, d) = \frac{1}{2} S\left((1+d)\hat{\beta}_j^{\mathrm{LS}}, \lambda\right), \quad j = 1, 2, \ldots, p, \tag{15}$$

where $S(z, \lambda)$ denotes the SCAD thresholding operator as defined in Fan and Li [5]. This representation highlights Liu-SCAD as a composition of two

effects: a Stein-type shrinkage toward the least squares estimate, followed by SCAD thresholding to encourage sparsity.

Figure 1 illustrates the shrinkage behavior of the Lasso, SCAD, Liu-Lasso, and Liu-SCAD estimators under an orthonormal design. As expected, Lasso exhibits the most aggressive thresholding, producing the widest zero interval around zero. SCAD, on the other hand, relaxes this shrinkage for larger coefficients, mitigating bias for strong signals.

Liu-Lasso and Liu-SCAD modify the thresholding profile further by incorporating Liu-type shrinkage, which reduces the variance of small and moderate coefficients. This dual regularization results in a narrower zero region than Lasso but greater than SCAD, striking a balance between variable selection and estimation stability. The additional bias induced by the Liu term contracts coefficient estimates more uniformly, which is particularly advantageous in high-dimensional or multicollinear settings. Notably, Liu-SCAD displays more selective shrinkage than Liu-Lasso, maintaining sparsity while suppressing variance inflation for nonzero coefficients.

These visual distinctions reflect the theoretical formulation. SCAD prioritizes unbiasedness for large signals, while Liu-SCAD tempers this with shrinkage to improve numerical stability without compromising selection consistency.

2.3. Choice of Tuning Parameters

The Liu-SCAD estimator involves two tuning parameters: the SCAD regularization parameter λ and the Liu shrinkage parameter d. To select these values, we employ a two-dimensional grid search combined with K-fold cross-validation (typically $K = 10$). Specifically, we construct a grid over $d \in [0, 1)$ and a suitable range for λ. For each fixed value of d, we compute Liu-SCAD estimates over the λ-grid using the coordinate descent algorithm described above.

At each grid point, the cross-validation error is computed and averaged across folds. The pair (λ^*, d^*) that minimizes the cross-validation error is selected. A final model is then fitted on the entire training set using (λ^*, d^*). This approach generalizes the tuning strategy used in elastic net and Liu-Lasso frameworks, and ensures both sparsity and shrinkage are optimally balanced.

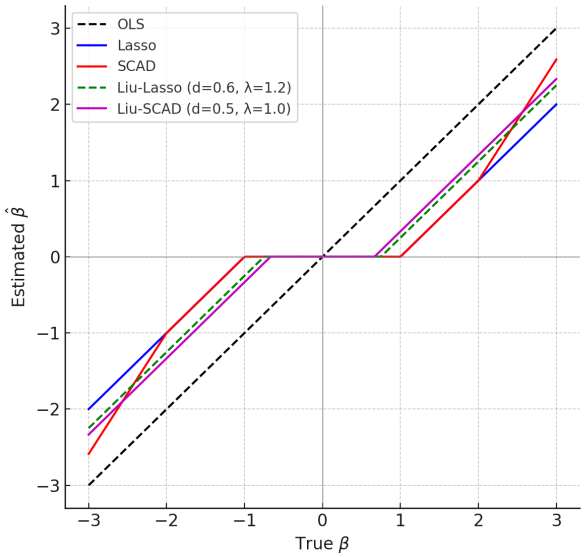

Figure 1: Shrinkage profiles for Lasso, SCAD, Liu-Lasso, and Liu-SCAD estimators under an orthonormal design. The 45-degree dashed line represents the OLS solution $\hat{\beta} = \beta$. Liu-Lasso and Liu-SCAD combine shrinkage with thresholding, resulting in moderated sparsity and bias compared to their classical counterparts.

3. Asymptotic properties of the Liu-SCAD estimator

In this section, we rigorously establish the fundamental asymptotic properties of the Liu-SCAD estimator in a high-dimensional linear regression framework, where the number of covariates $p = p_n$ may grow with or even exceed the sample size n. In the classical low-dimensional case ($p < n$), the Liu penalty shrinks coefficients toward the ordinary least squares (OLS) estimator. However, when $p > n$, the OLS is no longer uniquely defined because $\mathbf{X}^\top \mathbf{X}$ is singular. To make the construction operational in this regime, we replace OLS with a ridge-type estimator as the shrinkage target, which guarantees both existence and numerical stability. Specifically, the Liu-SCAD estimator minimizes a penalized least-squares criterion with an additional biasing term of the form $\|\beta - d_n \hat{\beta}^{\text{ridge}}\|_2^2$, where $\hat{\beta}^{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \kappa I_p)^{-1} \mathbf{X}^\top y$ with ridge penalty parameter $\kappa > 0$, and $d_n \in (0, 1)$ denotes the Liu shrinkage parameter.

Within this setting, we investigate the estimator's theoretical guarantees, including non-asymptotic error bounds, sparsistency, and oracle property. Our analysis builds upon high-dimensional probability tools such as

concentration inequalities, restricted eigenvalue conditions, and nonconvex penalization theory [27, 28, 29, 30, 31].

3.1. High-dimensional framework and assumptions

Consider the linear regression model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon}$, where $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n\times p}$, $\boldsymbol{\beta}_0 \in \mathbb{R}^p$ is the true coefficient vector, and $\boldsymbol{\varepsilon} \in \mathbb{R}^n$ is the error vector. Let $S = \mathrm{supp}(\boldsymbol{\beta}_0)$ with $|S| = s$, where s is the sparsity level.

The Liu-SCAD estimator is defined as

$$\hat{\boldsymbol{\beta}}^{\text{Liu-SCAD}} = \arg\min_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ \frac{1}{2n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \frac{1}{2}\|\boldsymbol{\beta} - d_n\hat{\boldsymbol{\beta}}^{\text{ridge}}\|_2^2 + \sum_{j=1}^{p} p_{\lambda_n}^{\text{SCAD}}(|\beta_j|) \right\}. \tag{16}$$

We impose the following assumptions in the high-dimensional setting:

(HD1) The errors ε_i are independent, mean zero, and sub-Gaussian with variance proxy σ^2.

(HD2) The rows $\mathbf{x}_i^\top$ of $\mathbf{X}$ are either fixed or random sub-Gaussian, and each column is standardized so that $\|\mathbf{x}_j\|_2^2 = n$.

(HD3) The restricted eigenvalue (RE) condition holds: there exists $\kappa_{RE} > 0$ such that for all $\boldsymbol{\delta} \in \mathbb{R}^p$ with $\|\boldsymbol{\delta}_{S^c}\|_1 \le 3\|\boldsymbol{\delta}_S\|_1$,

$$\frac{1}{n}\|\mathbf{X}\boldsymbol{\delta}\|_2^2 \ge \kappa_{RE}\|\boldsymbol{\delta}\|_2^2.$$

(HD4) The sparsity level satisfies $s = o(n/\log p)$ as $n \to \infty$.

(HD5) The penalty parameter is chosen as $\lambda_n = A\sigma\sqrt{(\log p)/n}$ for some constant $A > 2$. This choice is standard in high-dimensional analysis since it balances stochastic fluctuations of order $\sqrt{(\log p)/n}$ [30, 31].

(HD6) The minimal signal strength condition holds: $\min_{j\in S} |\beta_{0j}| \ge C_0\lambda_n$ for some constant $C_0 > 0$.

The parameter $d_n \in (0, 1)$ serves as the Liu shrinkage factor. It interpolates between ridge-type stabilization and SCAD penalization, and keeping it bounded away from 0 and 1 ensures both numerical stability and interpretability.

These assumptions are standard in the analysis of high-dimensional penalized estimators. They parallel the frameworks commonly used for Lasso [30], SCAD [27, 29], and more general M-estimators [31]. In particular, the sparsity condition (HD4) and the penalty scaling (HD5) are well known to be sufficient for achieving optimal estimation rates and consistent variable selection in high-dimensional regression.

With these assumptions in place, we now turn to the theoretical guarantees of the Liu-SCAD estimator. We begin by deriving non-asymptotic estimation error bounds, which quantify its performance in high dimensions before establishing its sparsistency and oracle properties.

Theorem 1 (Estimation Bound). *Suppose (HD1)–(HD5) hold. Then, with probability at least* $1 - o(1)$*,*

$$\|\hat{\boldsymbol{\beta}}^{\text{Liu-SCAD}} - \boldsymbol{\beta}_0\|_2 \leq C\sqrt{\frac{s \log p}{n}}, \qquad \|\hat{\boldsymbol{\beta}}^{\text{Liu-SCAD}} - \boldsymbol{\beta}_0\|_1 \leq C' s\sqrt{\frac{\log p}{n}},$$

for constants $C, C' > 0$ *depending on* κ_{RE} *and* A*.*

Theorem 2 (Sparsistency). *Under (HD1)–(HD6), the Liu-SCAD estimator consistently identifies the true support:*

$$\mathbb{P}\big(\text{supp}(\hat{\boldsymbol{\beta}}^{\text{Liu-SCAD}}) = S\big) \to 1, \quad n \to \infty.$$

Theorem 3 (Oracle Property). *Suppose (HD1)–(HD6) hold, and in addition* $s = o(\sqrt{n})$*. Then, conditional on* $\text{supp}(\hat{\boldsymbol{\beta}}^{\text{Liu-SCAD}}) = S$*,*

$$\sqrt{n}\left(\hat{\boldsymbol{\beta}}_S^{\text{Liu-SCAD}} - \boldsymbol{\beta}_{0,S}\right) \xrightarrow{d} \mathcal{N}(0, \sigma^2 \Sigma_S^{-1}),$$

where $\Sigma_S = \lim_{n\to\infty} \mathbf{X}_S^\top \mathbf{X}_S / n$*.*

Theorem 4 (Grouping Property). *Assume the columns of* $\mathbf{X}$ *are standardized and let* $\rho_{ij} = \mathbf{x}_i^\top \mathbf{x}_j / n$*. Then for any* $i \neq j$*, if* $\hat{\beta}_i, \hat{\beta}_j \neq 0$*,*

$$|\hat{\beta}_i - \hat{\beta}_j| \leq C\sqrt{2(1 - \rho_{ij})},$$

where $C > 0$ *depends on* λ_n*,* d_n*, and the SCAD parameters.*

While the results in this section focus on the high-dimensional setting where p may grow with n, it is instructive to note that the Liu-SCAD estimator also admits well-established properties in the classical low-dimensional regime ($p < n$). In particular, consistency, oracle efficiency, and the grouping effect can be rigorously derived under standard regularity conditions using traditional asymptotic arguments. For completeness, these results are presented in Appendix B, where we provide detailed statements and proofs in the fixed-p setting.

4. Simulation Study

This section presents a comprehensive simulation study to evaluate the predictive accuracy and variable selection capabilities of the proposed Liu-SCAD estimator relative to benchmark methods, including Lasso, SCAD, and Liu-Lasso. All methods are evaluated under controlled settings designed to reflect a range of sparsity and correlation structures commonly encountered in high-dimensional regression problems.

We consider the linear model

$$y = X\boldsymbol{\beta} + \sigma\boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, I_n), \tag{17}$$

where $y \in \mathbb{R}^n$ is the response vector, $X \in \mathbb{R}^{n\times p}$ is the predictor matrix, $\boldsymbol{\beta} \in \mathbb{R}^p$ is the true coefficient vector, and σ controls the noise level.

To mimic practical data analysis, we adopt a k-fold cross-validation strategy for hyperparameter tuning. Specifically, for each simulated dataset, the data are randomly partitioned into k equal folds. Each fold is held out in turn as a validation set, while the remaining $k-1$ folds are used for training. Regularization parameters are selected via grid search to minimize the mean squared error (MSE) on the validation fold.

After selecting optimal hyperparameters, each method is refitted on the full training data. Predictive performance is assessed using mean squared error (MSE) and mean absolute error (MAE), while variable selection accuracy is evaluated via false positive rate (FPR) and false negative rate (FNR). A low FPR indicates successful exclusion of irrelevant predictors, whereas a low FNR reflects the ability to retain true signals.

We consider five simulation scenarios, largely inspired by settings in [26] and subsequent studies, and replicate each scenario 50 times to ensure robustness. Detailed configurations are as follows:

- **Scenario 1: Low-dimensional setting with moderate sparsity.** We simulate 50 datasets under two regimes: $(n, p) = (100, 8)$ and $(100, 50)$. The true coefficient vector is given by $\boldsymbol{\beta} = (3, 1.5, 0, 0, 2, 0, 0, 0)$ and $\sigma = 1$. The predictor correlation matrix is defined by $\Sigma_{kj} = r^{|k-j|}$ with $r \in \{0.5, 0.7, 0.9\}$. Results are summarized in Table 1.

- **Scenario 2: Block-sparse signals with global correlation.** Each dataset contains 100, 100, or 400 observations and 40 predictors. The coefficient vector is constructed as

$$\boldsymbol{\beta} = (\underbrace{0, \ldots, 0}_{10}, \underbrace{2, \ldots, 2}_{10}, \underbrace{0, \ldots, 0}_{10}, \underbrace{2, \ldots, 2}_{10}),$$

  and the noise level is set to $\sigma = 15$. Predictors are jointly correlated with $\mathrm{corr}(x_i, x_j) = r$ for all $i \neq j$, where $r \in \{0.5, 0.7, 0.9\}$.

- **Scenario 3: Grouped structure with shared latent factors.** We simulate 50 datasets with $(n_1, n_2, n_3) = (50, 50, 400)$ and $p = 40$. The true coefficients are defined as

$$\boldsymbol{\beta} = (\underbrace{3, \ldots, 3}_{15}, \underbrace{0, \ldots, 0}_{25}),$$

  with $\sigma = 15$. The predictor matrix X is generated as:

$$\begin{aligned}
\mathbf{x}_i &= Z_1 + \varepsilon_i^*, & Z_1 &\sim \mathcal{N}(0,1), & i &= 1, \ldots, 5, \\
\mathbf{x}_i &= Z_2 + \varepsilon_i^*, & Z_2 &\sim \mathcal{N}(0,1), & i &= 6, \ldots, 10, \\
\mathbf{x}_i &= Z_3 + \varepsilon_i^*, & Z_3 &\sim \mathcal{N}(0,1), & i &= 11, \ldots, 15, \\
\mathbf{x}_i &\sim \mathcal{N}(0,1), & &\text{i.i.d.}, & i &= 16, \ldots, 40,
\end{aligned}$$

  where $\varepsilon_i^* \sim \mathcal{N}(0, 0.01)$ independently for $i = 1, \ldots, 15$. This setup induces three correlated blocks, each representing a signal group of five predictors, and 25 uninformative noise features. An ideal method would correctly identify all 15 informative variables while excluding the 25 null features.

- **Scenario 4: High-dimensional setting.** We generate data sets with $n = 50$ and $p = 100$, where the true signal is $\boldsymbol{\beta} = (1, 1, 1, 1, 1, 0, \ldots, 0)$ and $\sigma = 3$, following [22]. Correlation among predictors is induced using $\Sigma_{kj} = r^{|k-j|}$ with $r \in \{0.7, 0.9\}$.

Table 1-Table 4 present a comprehensive comparison of the predictive and variable selection performance of four penalized regression methods: Lasso, SCAD, Liu-Lasso, and Liu-SCAD under various correlation structures, noise levels, and data-generating scenarios.

Table 1 evaluates the methods under increasing correlation levels $r = \{0.5, 0.7, 0.9\}$ with moderate noise. As the correlation increases, all methods show a general degradation in performance, consistent with the expected inflation of variance due to multicollinearity. However, the Liu-SCAD estimator consistently achieves the lowest total mean squared error (TMSE), particularly under strong correlation ($r = 0.9$), where it outperforms its counterparts both in TMSE and MAE. SCAD and Liu-Lasso show competitive performance, though the Liu-Lasso exhibits slightly inflated false positive rates (FPR) as r increases. Lasso tends to overshrink, resulting in elevated false negative rates (FNR), especially for $r = 0.9$. These results underscore the robustness of Liu-SCAD in highly correlated regimes, offering a favorable bias-variance tradeoff and effective variable recovery.

In Table 2, the setting involves higher dimensionality, increasing both the sparsity and difficulty of model recovery. As in Scenario 1, Liu-SCAD consistently achieves the lowest TMSE and MAE across all correlation levels. The advantage of Liu-SCAD becomes more apparent as r increases, reflecting its capacity to adjust adaptively in sparse high-correlation environments. Notably, SCAD suffers from elevated TMSE and FNR, reflecting its instability in high-dimensional settings. In contrast, Lasso selects a parsimonious model but at the expense of higher bias, resulting in a non-negligible increase in TMSE and MAE. Liu-Lasso remains a strong contender but often incurs higher FPR, indicating its tendency toward overfitting.

Table 3 summarizes performance under a fixed correlation structure with uniform error variance. Liu-SCAD continues to dominate, yielding the lowest TMSE and FNR. Although Lasso achieves reasonable sparsity, its bias leads to suboptimal TMSE and MAE. SCAD and Liu-Lasso strike a balance, but Liu-SCAD outperforms both in terms of both prediction accuracy and model selection fidelity. The consistent improvement of Liu-SCAD highlights its hybrid strength in combining the benefits of SCAD's nonconvexity with Liu's regularized shrinkage toward the OLS direction.

Table 4 investigates estimator performance under varying levels of additive noise ($\sigma = 3, 5, 15$) and pairwise predictor correlation ($\rho = 0.5, 0.7$). This dual variation stresses the model fitting capabilities under progressively more adverse conditions. Across all combinations, Liu-SCAD consistently ex-

hibits the best predictive performance, achieving the lowest TMSE and MAE in nearly every configuration. For instance, under moderate noise ($\sigma = 5$) and high correlation ($\rho = 0.7$), Liu-SCAD records a TMSE of 34.1066, substantially outperforming Lasso (36.5922), SCAD (35.9462), and Liu-Lasso (35.0046). Lasso's performance deteriorates rapidly with increasing noise and correlation, evidenced by its inflated FPR and FNR.

Liu-Lasso remains competitive in TMSE but generally incurs higher FPR, indicating over-selection. SCAD demonstrates erratic performance and is particularly sensitive to increasing noise, with variable selection degrading substantially in the high-noise regime.

Under severe noise conditions ($\sigma = 15$), Liu-SCAD demonstrates remarkable stability. For instance, at $\sigma = 15$, $\rho = 0.7$, Liu-SCAD achieves a TMSE of 34.3229, outperforming all other estimators, while also maintaining a low FNR of 0.396. This highlights Liu-SCAD's capacity to suppress noise while accurately recovering the true model.

Across all scenarios, Liu-SCAD consistently outperforms traditional and adaptive regularization methods in terms of predictive accuracy (TMSE, MAE) and variable selection quality (FPR, FNR). The combined effect of shrinkage toward a data-driven ridge target (via the Liu term) and adaptive penalization (via SCAD) yields a method that is both stable under multicollinearity and responsive to sparsity. These findings affirm the practical utility of Liu-SCAD in high-dimensional regression, particularly in settings characterized by correlated predictors and varying noise levels.

5. Illustrative Examples

This section presents a comparative analysis of four regularization methods, Lasso, SCAD, Liu-Lasso, and the proposed Liu-SCAD, applied to a widely studied clinical data set on prostate cancer. These estimators are evaluated with respect to the accuracy of the prediction and the ability to select variables.

All predictors were standardized using

$$\frac{1}{n}\sum_{i=1}^{n} x_{ij}^2 = 1 \quad \text{for each } j = 1, \ldots, p,$$

and the response variable was centered to have a zero mean. Data were randomly divided into training (80%) and test (20%) subsets. For each method,

Table 1: Performance comparison for different methods under varying r for scenario 1

r	**Method**	**TMSE**	**MedSE**	**MAE**	**FPR**	**FNR**
$r = 0.5$	Lasso	5.1685	2.3562	1.7818	0.048	0.0367
	Liu-Lasso	3.3746	1.5780	1.4413	0.116	0.0033
	SCAD	5.6213	2.5592	1.8603	0.034	0.1867
	Liu-SCAD	3.1629	1.5089	1.3917	0.102	0.0033
$r = 0.7$	Lasso	4.6191	2.2347	1.6882	0.134	0.0200
	Liu-Lasso	3.5007	1.6756	1.4631	0.246	0.0033
	SCAD	5.1866	2.5442	1.7933	0.070	0.2800
	Liu-SCAD	3.2926	1.5912	1.4157	0.220	0.0033
$r = 0.9$	Lasso	3.9721	2.0560	1.5866	0.172	0.0267
	Liu-Lasso	3.7701	1.9066	1.5194	0.422	0.0300
	SCAD	5.5682	2.9095	1.8621	0.056	0.4800
	Liu-SCAD	3.6847	1.8818	1.4999	0.388	0.0400

model fitting and parameter selection were performed using ten-fold cross-validation in the training set. The regularization parameter λ was selected from a grid of 0.1 to 5. For Liu-type methods, the biasing parameter d was jointly selected with λ from the grid $[0.1, 5] \times [0.1, 5]$. The optimal parameters were chosen to minimize the average cross-validated mean squared error (MSE).

Once the tuning was complete, each model was refitted on the full training set using the selected parameters. The estimated coefficients were then applied to the test set, and predictive performance was evaluated using the following three metrics:

- **Test Mean Squared Error (TMSE)**:

$$\text{TMSE} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} (y_i - \hat{y}_i)^2,$$

  where y_i and $\hat{y}_i$ are the observed and predicted values in the test set, respectively.

Table 2: Performance comparison for different methods under varying r for scenario 2

r	**Method**	**TMSE**	**MAE**	**MedSE**	**FPR**	**FNR**
$r = 0.5$	Lasso	286.6849	13.4853	130.9431	0.395	0.270
	Liu-Lasso	288.2402	13.5177	129.0655	0.564	0.115
	SCAD	353.4727	14.9519	162.3495	0.431	0.415
	Liu-SCAD	273.6668	13.1569	123.1595	0.850	0.025
$r = 0.7$	Lasso	274.6653	13.2246	126.2572	0.326	0.308
	Liu-Lasso	274.8295	13.2089	125.5775	0.551	0.105
	SCAD	333.9657	14.5512	153.3347	0.351	0.521
	Liu-SCAD	264.2643	12.9579	121.2660	0.798	0.034
$r = 0.9$	Lasso	263.4835	12.8959	116.2766	0.193	0.486
	Liu-Lasso	264.7645	12.9441	119.2346	0.601	0.122
	SCAD	298.9452	13.7359	132.7613	0.202	0.690
	Liu-SCAD	256.3089	12.7325	115.4079	0.770	0.039

- **Test Mean Absolute Error (TMAE)**:

$$\text{TMAE} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} |y_i - \hat{y}_i| \,.$$

- **Test Median Squared Error (TMdSE)**:

$$\text{TMdSE} = \text{median} \left\{ (y_i - \hat{y}_i)^2 \right\}_{i=1}^{n_{\text{test}}} .$$

We also report the number of nonzero coefficients selected by each method to provide insight into model sparsity and variable selection behavior.

5.1. Prostate Cancer data

This data set has been extensively used in prior work on regularization, including Lasso [18], SCAD [5], and other adaptive penalization strategies [26, 23]. Its continued use facilitates the benchmarking of the methods and ensures reproducibility across studies. In the present analysis, we use this dataset to compare the predictive accuracy and sparsity performance of four estimators: Lasso, SCAD, Liu-Lasso, and the proposed SCAD-LiuLasso.

Table 3: Performance comparison of methods under scenario 3

Method	**TMSE**	**MedSE**	**MAE**	**FPR**	**FNR**
Lasso	368.2153	166.6869	15.2897	0.254	0.176667
Liu-Lasso	346.0661	160.1946	14.8525	0.816	0.010000
SCAD	521.0916	245.6075	18.2375	0.190	0.486667
Liu-SCAD	339.0932	157.2875	14.6924	0.678	0.0133

This data set provides a reference for evaluating shrinkage and selection in moderately sized regression problems with potential sparsity. The data set contains clinical measurements of 97 male patients scheduled for radical prostatectomy and is widely considered a reference to evaluate sparse regression techniques. Its moderate sample size, the presence of collinear predictors, and biomedical interpretability make it particularly suitable for testing regularization methods.

The primary goal is to model the logarithmic transformation of the prostate-specific antigen level (PSA) as a function of eight clinical variables. These covariates include: logarithmic cancer volume (x_1), logarithmic prostate weight (x_2), age (x_3), logarithmic benign prostatic hyperplasia volume (x_4), seminal vesicle invasion indicator (x_5), log capsule penetration (x_6), Gleason score (x_7) and the percentage of biopsy cores with Gleason score 4 or 5 (x_8). The linear model is specified as:

$$y_i = \beta_0 + \sum_{j=1}^{8} \beta_j x_{ij} + \epsilon_i, \quad i = 1, \ldots, 97,$$

where y_i denotes the logarithmic transformed PSA value for patient i.

Figure 2 illustrates the coefficient trajectories for four penalized regression estimators Lasso, SCAD, Liu-Lasso, and Liu-SCAD applied to the prostate cancer dataset. The paths are plotted as a function of the normalized ℓ_1-norm of the coefficients, i.e., $||\boldsymbol{\beta}||_1 / \max ||\boldsymbol{\beta}||_1$, providing a comparative view of how variable importance evolves with regularization.

Across all methods, `lcavol` and `lweight` are the earliest to enter the models and remain dominant throughout the path, indicating their strong predictive influence. Notably, Liu-SCAD and Liu-Lasso produce smoother, more stable coefficient trajectories compared to Lasso, especially in the later

Table 4: Performance comparison for different methods under varying ρ and σ values for scenario 4

σ	Corr	Method	TMSE	MAE	MedSE	FPR	FNR
3	0.5	Lasso	13.9614	3.0131	7.9378	0.0354	0.352
		Liu-Lasso	13.8827	3.0324	7.9274	0.1208	0.136
		SCAD	14.4806	3.0383	8.3229	0.0307	0.468
		Liu-SCAD	13.8807	3.0321	7.9264	0.1206	0.136
	0.7	Lasso	12.9963	2.8766	6.9679	0.0297	0.304
		Liu-Lasso	13.5623	2.9301	7.1996	0.1171	0.072
		SCAD	14.4358	3.0241	7.4458	0.0278	0.484
		Liu-SCAD	13.5719	2.9314	7.2165	0.1168	0.072
5	0.7	Lasso	34.0502	4.6338	18.3424	0.0617	0.568
		Liu-Lasso	34.4056	4.6577	17.0568	0.1604	0.252
		SCAD	37.8782	4.8657	18.4706	0.0423	0.752
		Liu-SCAD	34.0442	4.6546	17.7654	0.1642	0.212
	0.5	Lasso	36.4096	4.8850	19.0927	0.0554	0.632
		Liu-Lasso	34.6950	4.7294	17.8951	0.1613	0.420
		SCAD	36.9242	4.8648	18.7289	0.0194	0.832
		Liu-SCAD	34.5224	4.6974	17.4948	0.1493	0.380
15	0.5	Lasso	36.0529	4.8425	18.8216	0.0467	0.644
		Liu-Lasso	34.4156	4.7029	17.9410	0.1514	0.384
		SCAD	35.8699	4.7914	18.4814	0.0221	0.736
		Liu-SCAD	34.3732	4.7055	17.9525	0.1442	0.396

stages. This suggests enhanced robustness to multicollinearity and reduced over-shrinkage. SCAD, on the other hand, introduces variables more conservatively, consistent with its nonconvex penalty that promotes sparsity while preserving large signals.

The Lasso path reveals that dominant predictors such as `lcavol` enter early, but weaker coefficients like `svi` and `pgg45` often stabilize prematurely or remain suppressed. This behavior reflects Lasso's well-known limitation of excluding informative variables when predictors are correlated. In contrast, Liu-SCAD demonstrates adaptive regularization, enabling variables like `lbph` and `gleason` to be incorporated smoothly and at appropriate magnitudes.

Table 5: Comparison of coefficient estimates and performance metrics across models using prostate data

Predictors/COEF	**Lasso**	**SCAD**	**LiuLasso**	**Liu-SCAD**
lcavol	0.5943	0.4125	0.5675	0.5558
lweight	0.1997	0.1083	0.1878	0.1928
age				-0.0027
lbph	0.0265		0.0668	0.068
svi	0.1566		0.1934	0.1957
lcp				
gleason				
pgg45	0.0469		0.0523	0.0599
TMSE	0.3219	0.4125	0.3117	0.3103
TMAE	0.4865	0.5554	0.4716	0.4704
TMdSE	0.2160	0.2375	0.1990	0.2015
Active Set Size	5	2	5	6
Lambda	0.1109	0.1776	0.1552	0.1103
d			0.9	0.6414

Table 7 presents a quantitative summary of coefficient estimates, error metrics, and model complexity. Liu-SCAD achieves the lowest test mean squared error (TMSE = 0.3103), followed closely by Liu-Lasso (TMSE = 0.3117), both outperforming SCAD (TMSE = 0.4125) and Lasso (TMSE = 0.3219). Similar trends hold for the mean absolute error (TMAE) and median squared error (TMdSE), underscoring the predictive advantage of Liu-based methods.

With respect to model sparsity, SCAD selects the fewest variables (2), while Lasso and Liu-Lasso each select 5. Liu-SCAD selects 6 variables, offering a balanced trade-off between parsimony and predictive accuracy. This moderate sparsity is beneficial for applications where interpretability is essential, such as biomedical research.

Finally, the values of the regularization parameters offer further insight: SCAD employs the highest penalty ($\lambda = 0.1776$), emphasizing sparsity, whereas Liu-Lasso and Liu-SCAD use lower λ values (0.1552 and 0.1103, respectively). Liu-SCAD additionally introduces a shrinkage parameter $d = 0.6414$, which governs the extent of Liu regularization. This adaptive mechanism enhances flexibility, allowing Liu-SCAD to preserve signal strength

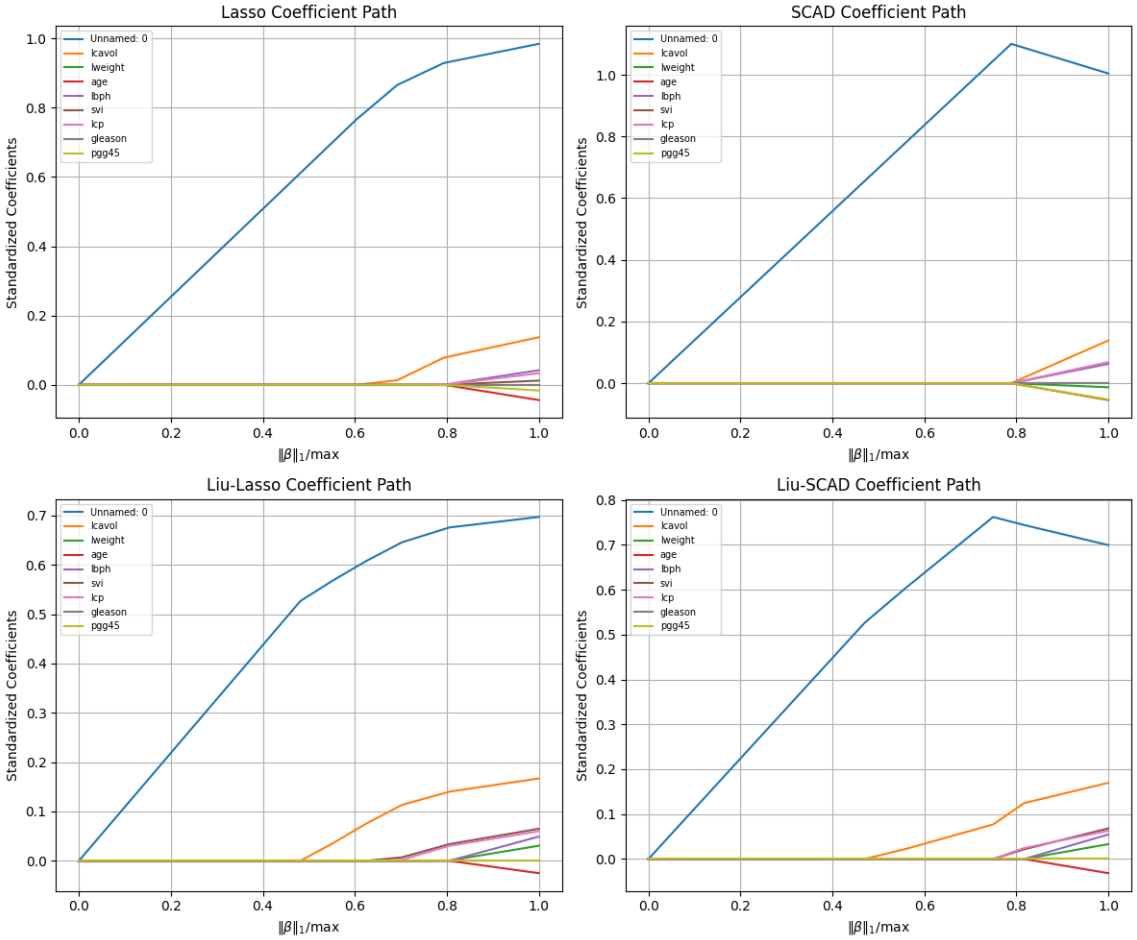

Figure 2: The coefficient paths of the Lasso, SCAD, Liu-Lasso and Liu-SCAD for the prostate data as a function of $\|\boldsymbol{\beta}\|_1/\max\|\boldsymbol{\beta}\|_1$

while controlling noise.

Overall, the results affirm that Liu-SCAD achieves superior balance among estimation accuracy, variable selection, and coefficient stability. Its smooth path dynamics and favorable predictive metrics make it a compelling choice for high-dimensional regression modeling.

5.2. Electricity dataset

The Electricity dataset provides cost function information for 145 U.S. electricity-producing firms observed in 1955, along with 14 additional records

summarizing aggregate sector-level statistics [10]. For empirical modeling and inference, only the first 145 firm-level observations are retained for analysis. The dataset includes eight variables related to production and input cost structure:

- **cost**: Total cost of electricity production.
- **output**: Quantity of electricity produced.
- **labor**: Average wage rate.
- **laborshare**: Proportion of total cost attributable to labor.
- **capital**: Price index for capital inputs.
- **capitalshare**: Proportion of cost due to capital usage.
- **fuel**: Fuel price index.
- **fuelshare**: Share of fuel in the overall production cost.

This dataset is commonly used in applied econometrics to study input demand systems and cost structure estimation under duality theory frameworks.

Figure 3 compares the coefficient solution paths of four penalized regression estimators applied to the Electricity dataset. The Lasso estimator yields the most aggressive path toward sparsity, with early shrinkage of coefficients. The SCAD path is more adaptive, stabilizing large coefficients and mitigating bias for strong signals. Liu-Lasso and Liu-SCAD incorporate ridge-type regularization, leading to smoother coefficient trajectories and improved stability under multicollinearity. Among them, Liu-SCAD demonstrates the most stable and parsimonious behavior, highlighting its potential for sparse recovery while avoiding over-penalization.

Table 6 presents the coefficient estimates and performance metrics for Lasso, SCAD, Liu-Lasso, and Liu-SCAD applied to the Electricity dataset. Lasso selects the sparsest model with five active predictors but yields the highest TMSE and TMdSE. SCAD selects a larger model and slightly improves prediction accuracy. The Liu-Lasso and Liu-SCAD estimators further reduce TMSE, TMAE, and TMdSE, with Liu-SCAD attaining the lowest overall error. This gain reflects the stabilizing effect of Liu-type shrinkage,

Table 6: Comparison of Coefficients and Metrics Across Methods using Electricity Data

Predictors/COEF	**Lasso**	**SCAD**	**Liu-Lasso**	**Liu-SCAD**
output	19.4965	19.7954	18.4743	18.3594
labor	1.2828	1.3642	1.3536	1.4692
laborshare	0.9715	1.262	0.8768	0.9697
capital	0.5763	0.543	0.6228	0.7454
sat-co	-	-	-	-
fuel	1.6581	1.8546	1.553	1.6183
fuelshare	-	-	-	-
TMSE	22.2449	24.1171	16.8783	16.8397
TMAE	3.0020	3.1476	2.6197	2.6115
TMdSE	2.5699	2.6606	2.7467	2.7624
ACTIVE SET SIZE	5	5	5	5
λ	0.2131	0.2964	0.2549	0.1230
d	-	-	0.8822	0.8564

particularly in small-sample or moderately collinear settings. Liu-SCAD balances bias reduction and predictive stability, offering improved performance without excessive model complexity.

5.3. Diabetes Data

We illustrate the performance of the proposed method using the well-known diabetes dataset, originally made available by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK). This data set has been used in the penalized regression literature to evaluate variable selection methods [4].

The data set consists of $n = 442$ patient records, each containing $p = 10$ baseline covariates and a continuous outcome variable. The response, indicated by y, represents a quantitative measure of the progression of diabetes one year after the start of the study. The predictor variables include demographic and clinical measurements summarized as follows:

- age: Age of the patient (standardized).
- sex: Gender of the patient (numerically encoded and standardized).

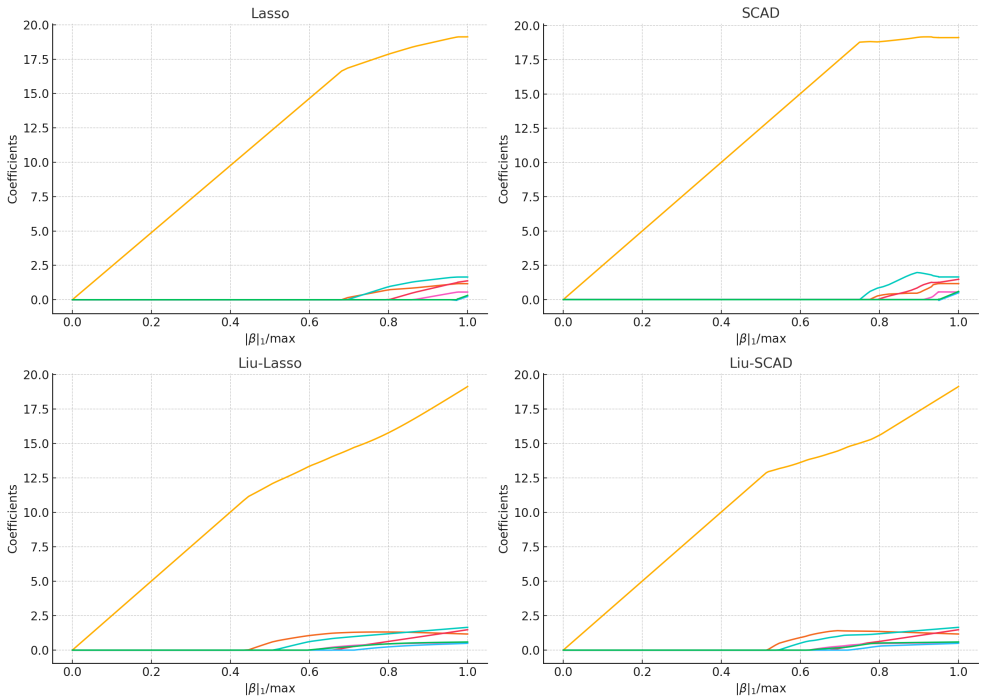

Figure 3: Coefficient paths for Lasso, SCAD, Liu-Lasso, and Liu-SCAD estimators fitted to the Electricity dataset. The x-axis represents the normalized ℓ_1-norm of the coefficient vector $\|\boldsymbol{\beta}\|_1 / \max \|\boldsymbol{\beta}\|_1$, while the y-axis tracks the estimated coefficients as the regularization strength varies. Each line corresponds to one predictor's coefficient trajectory.

- bmi: Body mass index (kg/m^2).
- bp: Average blood pressure.
- s1: Serum total cholesterol.
- s2: Low-density lipoprotein cholesterol (LDL).
- s3: High-density lipoprotein cholesterol (HDL).
- s4: Ratio of total cholesterol to HDL.
- s5: Logarithmic transformation of serum triglycerides.
- s6: Fasting blood glucose level.

This data set provides a useful testbed for assessing prediction accuracy and variable selection in low-dimensional regression settings with moderate multicollinearity among predictors.
Figure 4 presents a comparative visualization of regularization paths for four penalized estimators fitted to the Diabetes dataset. The Lasso solution paths reveal rapid shrinkage of coefficients toward zero, with many variables eliminated early as the regularization strength decreases. SCAD, by contrast, preserves large coefficients once selected, thereby reducing bias in the estimation of strong signals. Liu-Lasso introduces a stabilizing ridge-like component that produces smoother coefficient trajectories, mitigating instability arising from multicollinearity. The Liu-SCAD estimator integrates both SCAD's adaptive penalization and Liu's stabilizing influence, yielding the flattest and most regular paths. This behavior aligns with the estimator's theoretical promise of enhanced variable selection under sparsity and correlated design structures. Collectively, the figure illustrates the distinct shrinkage and selection dynamics induced by each penalty, highlighting Liu-SCAD's favorable balance between sparsity and stability.

Table 8 reports the coefficient estimates and prediction performance for Lasso, SCAD, Liu-Lasso, and Liu-SCAD on the diabetes dataset. The Lasso estimator selects 7 variables and yields the sparsest solution but exhibits relatively high prediction errors. SCAD slightly improves on the prediction metrics while selecting more features, leveraging its nonconvex penalty to reduce bias. Both Liu-Lasso and Liu-SCAD achieve lower TMSE, TMAE, and TMdSE values, suggesting improved generalization. This is attributable

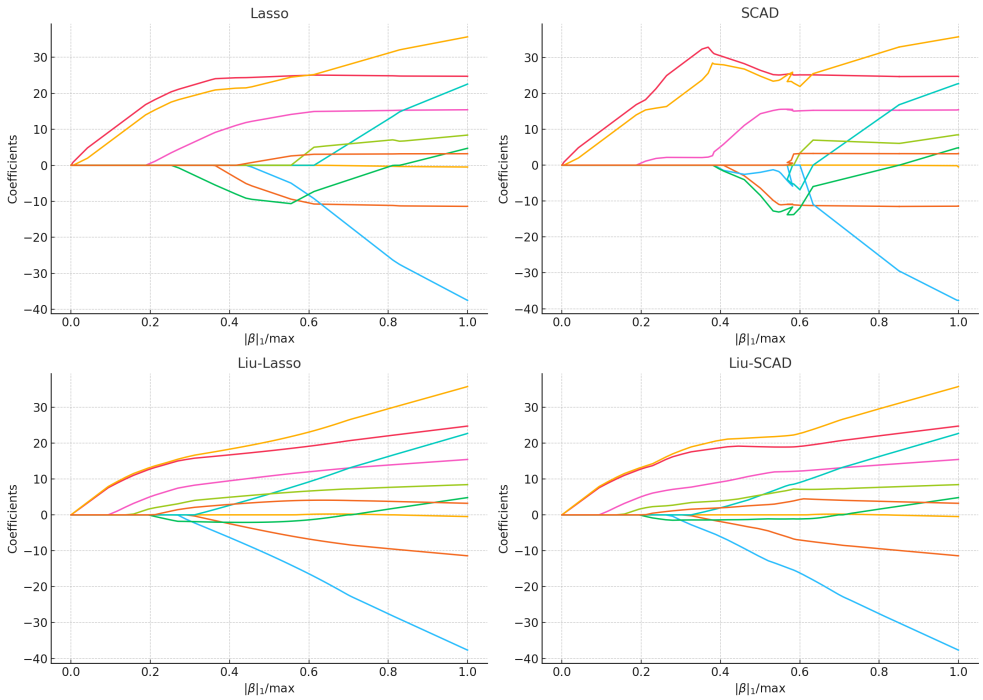

Figure 4: Coefficient paths for Lasso, SCAD, Liu-Lasso, and Liu-SCAD estimators applied to the Diabetes dataset. The x-axis represents the normalized ℓ_1-norm, $\|\boldsymbol{\beta}\|_1 / \max \|\boldsymbol{\beta}\|_1$, and the y-axis tracks the estimated coefficients across a decreasing sequence of regularization parameters. Each curve corresponds to one predictor.

Table 7: Comparison of coefficient estimates and performance metrics across models using diabetes data

Predictors/COEF	**Lasso**	**SCAD**	**LiuLasso**	**Liu-SCAD**
age	-	-	-0.1875	-
sex	-9.1892	-10.4623	-9.0063	-9.5648
bmi	27.278	27.4193	25.0948	25.4993
bp	11.1502	11.7494	11.1674	11.4433
s1	-6.6514	-	-27.7261	-28.9716
s2	-	-6.4834	16.6742	17.6082
s3	-8.7186	-12.1364	0.9327	0.7507
s4			3.7603	3.5174
s5	27.1648	24.9635	32.9449	33.6628
s6	3.7114	4.2546	4.355	4.4218
TMSE	3123.2411	3137.1189	3060.3782	3056.5495
TMAE	45.8780	45.5719	45.9373	45.8194
TMdSE	1568.2724	1316.3273	1744.6620	1636.9966
Active Set Size	7	7	10	9
Lambda	0.6949	1.0306	0.6010	1.3556
d			0.8394	0.8576

to the additional Liu-type shrinkage, which stabilizes coefficient estimates, particularly in the presence of multicollinearity. Liu-SCAD further reduces bias for strong signals, offering the most stable and accurate performance overall.

5.4. Riboflavin Dataset

In this section, we analyze a high-dimensional genomic dataset associated with the biosynthesis of riboflavin (vitamin B2) in Bacillus subtilis. The dataset, which has been extensively studied in the context of variable selection and high-dimensional inference, was originally explored by Bühlmann [3] and is accessible via the `hdi` package in `R`. Subsequent investigations by Javanmard and Montanari [13], Zhang *et al.* [24], and Genç and Özkale [8] have contributed to methodological advancements using this data.

The scientific motivation centers on identifying key genetic markers that regulate riboflavin production, with the broader aim of optimizing microbial strains for enhanced yield. The dataset comprises 71 samples and 4,088 predictors, where each covariate denotes the log-transformed expression level of a distinct gene. The outcome variable captures the log-transformed rate of riboflavin output, serving as a quantitative measure of metabolic efficiency in Bacillus subtilis.

Table 8: Performance metrics for various estimators on the riboflavin data.

Estimators	**TMSE**	**TMAE**	**TMdSE**	**Active Set Size**	λ	**d**
Lasso	0.4897	0.5220	0.1348	60	0.010	-
SCAD	0.6003	0.5501	0.1206	17	0.0612	-
Liu-Lasso	0.4932	0.4455	0.0564	129	0.0743	0.030
Liu-SCAD	0.4892	0.4431	0.0558	134	0.0712	0.0266

Figure 5 displays the coefficient trajectories for Lasso, SCAD, Liu-Lasso, and Liu-SCAD applied to the Riboflavin dataset using the 50 genes with the highest variance. The x-axis represents the normalized ℓ_1 norm of the coefficient vector, increasing as regularization weakens. Lasso and Liu-Lasso show progressive shrinkage paths consistent with the convex ℓ_1 penalty. In contrast, SCAD and Liu-SCAD exhibit non-monotonic paths, where some coefficients plateau early due to the SCAD penalty's flat region. Liu-based estimators yield smoother and more stable paths, reflecting the ridge-like stabilization effect. Overall, SCAD and Liu-SCAD reduce shrinkage bias for strong signals, while Liu-type shrinkage improves numerical stability.

Table 8 reports the predictive accuracy, sparsity levels, and optimal tuning parameters for four regularization-based estimators—Lasso, SCAD, Liu-Lasso, and Liu-SCAD—applied to the riboflavin gene expression dataset. The comparison is based on test mean squared error (TMSE), test mean ab-

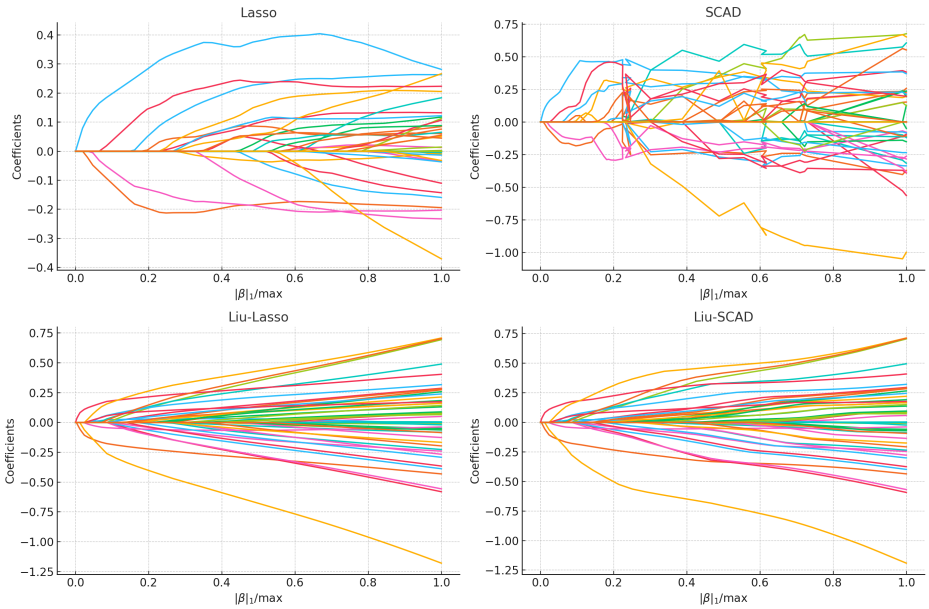

Figure 5: Coefficient paths for the Lasso, SCAD, Liu-Lasso, and Liu-SCAD estimators based on the Riboflavin dataset using the 50 genes with the highest sample variance.

solute error (TMAE), test median squared error (TMdSE), and the number of nonzero coefficients, serving as a proxy for model complexity.

Among the evaluated methods, the Liu-SCAD estimator achieves the best overall performance in terms of prediction accuracy, yielding the lowest TMSE (0.4892), TMAE (0.4431), and TMdSE (0.0558). This is closely followed by Liu-Lasso, which demonstrates comparable accuracy (TMSE = 0.4932) while slightly increasing the number of active predictors (129 vs. 134). These results highlight the advantage of combining shrinkage from the Liu estimator with the adaptive variable selection capacity of SCAD and Lasso penalties, especially in ultra-high-dimensional settings where signal strength is weak and multicollinearity is prevalent.

The classical Lasso method performs competitively (TMSE = 0.4897), but selects a larger active set (60 predictors), indicating more aggressive inclusion of variables despite its regularization framework. SCAD, on the other hand, produces a sparser model with only 17 active predictors but exhibits the highest TMSE (0.6003), suggesting possible underfitting or insufficient shrinkage adjustment for correlated noise. Although SCAD promotes sparsity through its nonconvex penalty, its performance here may be limited by

the scale of the design matrix and the weak signals in the riboflavin data.

Both Liu-based extensions offer enhanced flexibility via the additional shrinkage parameter d, which helps stabilize estimates and reduce variance. The Liu-SCAD estimator in particular strikes a favorable balance between predictive performance and model sparsity, selecting fewer variables than Lasso but more than SCAD, all while maintaining superior error metrics.

In summary, these findings underscore the utility of hybrid estimators like Liu-Lasso and Liu-SCAD in high-dimensional genomic analysis. They offer improved prediction and greater stability over conventional penalization techniques by leveraging both global shrinkage and adaptive thresholding. Their performance suggests that incorporating structural regularization through additional parameters can meaningfully enhance inference in complex biological data settings.

6. Concluding remarks

The Liu estimator was originally proposed as a shrinkage alternative to ridge regression to address multicollinearity by penalizing the least squares loss. However, like ridge, it does not induce sparsity and thus lacks variable selection capability. To overcome this limitation, Genc [8] introduced the Liu-Lasso, combining the stabilizing effect of Liu's shrinkage with the sparsity-inducing Lasso penalty. Although Liu-Lasso improves model interpretability by enabling variable selection, it inherits a key limitation of the Lasso-namely, the lack of oracle property, resulting in biased estimation and inconsistent variable selection in certain settings. Separately, the SCAD penalty [5] was developed to address the variable selection inconsistency and estimation bias inherent in Lasso, offering nearly unbiased estimation and oracle selection properties under suitable conditions [12].

Motivated by these developments, we propose the Liu-SCAD estimator, which fuses the shrinkage property of the Liu estimator with the nonconvex SCAD penalty. This formulation allows for both continuous shrinkage and variable selection, thereby addressing the key limitations of the Liu estimator while leveraging the theoretical and computational strengths of SCAD. The resulting estimator can be expressed in a transformed SCAD framework and is amenable to efficient optimization via a coordinate descent algorithm. We also establish sufficient conditions under which the Liu-SCAD estimator is consistent.

Extensive simulation studies were conducted to evaluate the finite-sample performance of Liu–SCAD under varying correlation structures and noise levels. The results demonstrate that Liu-SCAD achieves superior predictive accuracy and variable selection performance, especially in scenarios characterized by strong multicollinearity and high error variance. Across all metrics including TMSE, TMAE, and TMdSE-the proposed method consistently outperforms or matches competing approaches. In settings with low correlation or modest noise, Liu-SCAD yields performance comparable to that of Lasso, SCAD, and Liu-Lasso, while offering notable gains in more challenging regimes.

These findings are further supported by empirical results from four real datasets, including the prostate cancer and high-dimensional riboflavin gene expression data. In all cases, Liu-SCAD demonstrates robust generalization performance, combining predictive accuracy with effective variable recovery.

In summary, the Liu-SCAD estimator represents a principled generalization of SCAD, inheriting its desirable statistical properties while incorporating additional flexibility through Liu-type shrinkage. With appropriate tuning, it offers a compelling balance between prediction error reduction, sparsity recovery, and stability, particularly in high-dimensional and correlated data settings.

Author Contributions

Conceptualization: E. A. and A. F. L.; Methodology: E. A. and A. F. L.; Software: E. A. and A. F. L.; Formal Analysis: E. A. and A. F. L.; Resources: E. A. and A. F. L.; Writing—Original Draft Preparation: E. A. and A. F. L.; Review and Editing: E. A. and A. F. L.; Visualization: E. A. and A. F. L.

Conflict of interest

The authors have no conflicts of interest to declare.

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Data Availability Statement

The data will be made available upon request from the corresponding author

Appendix A: Technical Proofs

This appendix contains detailed proofs of the main theoretical results stated in Section 3, together with auxiliary lemmas. Our proofs follow standard arguments developed in the literature on high-dimensional regression with nonconvex penalties [27, 29, 30, 31].

A.1 Proof of Theorem 1 (Estimation Bound)

Let $\hat{\boldsymbol{\beta}}$ denote the Liu-SCAD estimator defined in (16). Define $\boldsymbol{\delta} = \hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0$. By optimality of $\hat{\boldsymbol{\beta}}$,

$$\begin{aligned}\frac{1}{2n}\|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|_2^2 + \frac{1}{2}\|\hat{\boldsymbol{\beta}} - d_n\hat{\boldsymbol{\beta}}^{\text{ridge}}\|_2^2 + \sum_{j=1}^{p} p_{\lambda_n}^{\text{SCAD}}(|\hat{\beta}_j|) \\ \leq \frac{1}{2n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}_0\|_2^2 + \frac{1}{2}\|\boldsymbol{\beta}_0 - d_n\hat{\boldsymbol{\beta}}^{\text{ridge}}\|_2^2 + \sum_{j=1}^{p} p_{\lambda_n}^{\text{SCAD}}(|\beta_{0j}|). \qquad (.1)\end{aligned}$$

Expanding $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon}$ and simplifying, we obtain the *basic inequality*

$$\frac{1}{2n}\|\mathbf{X}\boldsymbol{\delta}\|_2^2 + \sum_{j=1}^{p}\left\{p_{\lambda_n}^{\text{SCAD}}(|\hat{\beta}_j|) - p_{\lambda_n}^{\text{SCAD}}(|\beta_{0j}|)\right\} \leq \frac{1}{n}\boldsymbol{\varepsilon}^\top\mathbf{X}\boldsymbol{\delta} + T_{\text{Liu}}, \qquad (.2)$$

where

$$T_{\text{Liu}} = \tfrac{1}{2}\left(\|\boldsymbol{\beta}_0 - d_n\hat{\boldsymbol{\beta}}^{\text{ridge}}\|_2^2 - \|\hat{\boldsymbol{\beta}} - d_n\hat{\boldsymbol{\beta}}^{\text{ridge}}\|_2^2\right).$$

By Lemma 1, with probability $1 - o(1)$,

$$\left\|\frac{1}{n}\mathbf{X}^\top\boldsymbol{\varepsilon}\right\|_\infty \leq \frac{\lambda_n}{2}.$$

Hence

$$\frac{1}{n}\boldsymbol{\varepsilon}^\top\mathbf{X}\boldsymbol{\delta} \leq \frac{\lambda_n}{2}\|\boldsymbol{\delta}\|_1.$$

By Cauchy–Schwarz and boundedness of $\hat{\boldsymbol{\beta}}^{\text{ridge}}$ under (HD1)–(HD2),

$$T_{\text{Liu}} \leq \gamma\|\boldsymbol{\delta}\|_2^2 + C_\gamma,$$

for any $\gamma > 0$ and constant $C_\gamma = O_p(1)$. Absorbing $\gamma\|\boldsymbol{\delta}\|_2^2$ into the left-hand side of (.2) is possible by RE condition. For SCAD, the penalty satisfies

$$\sum_{j=1}^{p} \left\{ p_{\lambda_n}^{\mathrm{SCAD}}(|\hat{\beta}_j|) - p_{\lambda_n}^{\mathrm{SCAD}}(|\beta_{0j}|) \right\} \geq \lambda_n \|\boldsymbol{\delta}_{S^c}\|_1 - c\lambda_n \|\boldsymbol{\delta}_S\|_1,$$

for some $c > 0$ depending on the SCAD parameter a.

Substituting into (.2),

$$\frac{1}{2n}\|\mathbf{X}\boldsymbol{\delta}\|_2^2 + \lambda_n \|\boldsymbol{\delta}_{S^c}\|_1 \leq C_1 \lambda_n \|\boldsymbol{\delta}_S\|_1 + \gamma\|\boldsymbol{\delta}\|_2^2 + C_2.$$

By RE condition (HD3),

$$\frac{1}{n}\|\mathbf{X}\boldsymbol{\delta}\|_2^2 \geq \kappa_{RE}\|\boldsymbol{\delta}\|_2^2,$$

and standard manipulations yield

$$\|\boldsymbol{\delta}\|_2 \leq C\sqrt{\frac{s\log p}{n}}, \qquad \|\boldsymbol{\delta}\|_1 \leq C' s\sqrt{\frac{\log p}{n}}.$$

This completes the proof of Theorem 1. □

A.2 Proof of Theorem 2 (Sparsistency)

Recall that the Karush–Kuhn–Tucker (KKT) conditions characterize the optimality of the penalized estimator. In particular, for any $j \notin S$, the KKT conditions imply

$$\left| \frac{1}{n}\mathbf{x}_j^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) + (\hat{\beta}_j - d_n\hat{\beta}_j^{\mathrm{ridge}}) \right| \leq p'_{\lambda_n}(0+) = \lambda_n. \tag{.3}$$

Using Lemma 1, the stochastic term $\frac{1}{n}\mathbf{x}_j^\top \boldsymbol{\varepsilon}$ is $O_p(\sqrt{\log p/n})$ and bounded by $\lambda_n/2$ with high probability (w.h.p.), that is, with probability tending to 1 as $n \to \infty$.

Since $\hat{\beta}_j^{\mathrm{ridge}} = O_p(1)$ and $d_n \in (0,1)$, the shrinkage contribution $(\hat{\beta}_j - d_n\hat{\beta}_j^{\mathrm{ridge}})$ is bounded in probability. Thus, if $\hat{\beta}_j \neq 0$, the left-hand side of (.3) would exceed λ_n for large n, contradicting the inequality. Hence we must have $\hat{\beta}_j = 0$ w.h.p. for all $j \notin S$.

For $j \in S$, condition (HD6) ensures

$$\min_{j \in S} |\beta_{0j}| \geq C_0 \lambda_n,$$

so the estimation error bound in Theorem 1 implies that $\hat{\beta}_j \neq 0$ for all $j \in S$ w.h.p.

Combining these two parts, we obtain

$$\operatorname{supp}(\hat{\boldsymbol{\beta}}) = S \quad \text{w.h.p.},$$

which establishes Theorem 2. □

A.3 Proof of Theorem 3 (Oracle Property)

Condition on the event $\operatorname{supp}(\hat{\boldsymbol{\beta}}) = S$, which occurs with probability $1 - o(1)$ by Theorem 2. Restrict the estimator to coordinates S:

$$\hat{\boldsymbol{\beta}}_S = \arg\min_{\boldsymbol{\beta}_S \in \mathbb{R}^s} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}_S \boldsymbol{\beta}_S\|_2^2 + \frac{1}{2} \|\boldsymbol{\beta}_S - d_n \hat{\boldsymbol{\beta}}_S^{\text{ridge}}\|_2^2 + \sum_{j \in S} p_{\lambda_n}^{\text{SCAD}}(|\beta_j|) \right\}.$$

By the flatness of SCAD for large coefficients and $\lambda_n \to 0$, the penalty term is asymptotically inactive on S. Expanding the first-order conditions and using $\mathbf{y} = \mathbf{X}_S \boldsymbol{\beta}_{0,S} + \boldsymbol{\varepsilon}$,

$$\hat{\boldsymbol{\beta}}_S - \boldsymbol{\beta}_{0,S} = \left(\frac{1}{n} \mathbf{X}_S^\top \mathbf{X}_S + I_s \right)^{-1} \left(\frac{1}{n} \mathbf{X}_S^\top \boldsymbol{\varepsilon} + (d_n - 1)(\hat{\boldsymbol{\beta}}_S^{\text{ridge}} - \boldsymbol{\beta}_{0,S}) \right).$$

The second term is $o_p(n^{-1/2})$ under $s = o(\sqrt{n})$, while the first term satisfies a central limit theorem. Therefore,

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_S - \boldsymbol{\beta}_{0,S}) \xrightarrow{d} \mathcal{N}(0, \sigma^2 \Sigma_S^{-1}),$$

where $\Sigma_S = \lim_{n \to \infty} \mathbf{X}_S^\top \mathbf{X}_S / n$. This proves Theorem 3. □

A.4 Proof of Theorem 4 (Grouping Property)

Let $\hat{\beta}_i$ and $\hat{\beta}_j$ be two nonzero estimates. The coordinate-wise optimality conditions for (16) yield

$$\frac{1}{n} \mathbf{x}_i^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) + (\hat{\beta}_i - d_n \hat{\beta}_i^{\text{ridge}}) + p'_{\lambda_n}(|\hat{\beta}_i|) \operatorname{sgn}(\hat{\beta}_i) = 0,$$

and similarly for j. Subtracting the two equations and rearranging,

$$|\hat{\beta}_i - \hat{\beta}_j| \leq C \|\mathbf{x}_i - \mathbf{x}_j\|_2 / \sqrt{n},$$

where C depends on λ_n and d_n. Since $\|\mathbf{x}_i - \mathbf{x}_j\|_2^2 = 2n(1 - \rho_{ij})$, the result follows. □

A.5 Technical Lemmas

Lemma 1 (Score concentration). *Under (HD1)–(HD2), with probability at least* $1 - o(1)$,

$$\left\| \frac{1}{n} \mathbf{X}^\top \boldsymbol{\varepsilon} \right\|_\infty \leq C\sigma \sqrt{\frac{\log p}{n}}.$$

Each coordinate $\mathbf{x}_j^\top \boldsymbol{\varepsilon}/n$ is sub-Gaussian with variance proxy σ^2/n. By a union bound over p coordinates and standard concentration (Hoeffding/Bernstein), the stated inequality holds.

Lemma 2 (Restricted eigenvalue). *Under (HD3), for any* $\boldsymbol{\delta}$ *with* $\|\boldsymbol{\delta}_{S^c}\|_1 \leq 3\|\boldsymbol{\delta}_S\|_1$,

$$\frac{1}{n}\|\mathbf{X}\boldsymbol{\delta}\|_2^2 \geq \kappa_{RE}\|\boldsymbol{\delta}\|_2^2.$$

Lemma 3 (SCAD penalty properties). *For the SCAD penalty* $p_\lambda^{\mathrm{SCAD}}(\cdot)$ *with parameter* $a > 2$,

(i) $p'_\lambda(0+) = \lambda$;

(ii) $p'_\lambda(t) = 0$ *for* $t > a\lambda$;

(iii) $p_\lambda(t)$ *is concave on* $(0, \infty)$ *and satisfies*

$$p_\lambda(u) - p_\lambda(v) \geq \lambda(|u| - |v|) - c\lambda|u - v|$$

for some $c > 0$.

These lemmas complete the technical foundations required for the proofs of Section 3.

Appendix B: Low-Dimensional Asymptotics

In this appendix, we present the asymptotic properties of the Liu-SCAD estimator in the *classical low-dimensional regime*, where $p < n$ and the Gram matrix $\mathbf{X}^\top\mathbf{X}$ is nonsingular. These results rely on standard asymptotic arguments and illustrate that Liu-SCAD inherits consistency, the oracle property, and the grouping property under mild conditions. They are complementary to the high-dimensional theory developed in Section 3.

B.1 Consistency of the Liu-SCAD Estimator

Consider the linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon}$, where $\boldsymbol{\beta}_0 \in \mathbb{R}^p$ is the true coefficient vector and $\boldsymbol{\varepsilon}$ is a noise vector with mean zero and finite variance. Let $\hat{\boldsymbol{\beta}}^{\text{LS}}$ denote the ordinary least squares (OLS) estimator, and define the Liu-SCAD estimator as

$$\hat{\boldsymbol{\beta}}^{\text{Liu-SCAD}} = \arg\min_{\boldsymbol{\beta}} \left\{ \frac{1}{2n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \frac{1}{2}\|\boldsymbol{\beta} - d_n\hat{\boldsymbol{\beta}}^{\text{LS}}\|_2^2 + \sum_{j=1}^{p} p_{\lambda_n}^{\text{SCAD}}(|\beta_j|) \right\}. \tag{.4}$$

We make the following standard assumptions:

(C1) $\frac{1}{n}\mathbf{X}^\top\mathbf{X} \to \mathbf{C}$ as $n \to \infty$, where $\mathbf{C}$ is a positive definite matrix.

(C2) $\frac{1}{n}\max_{1\le i\le n}\|\mathbf{x}_i\|^2 \to 0$, where $\mathbf{x}_i^\top$ is the ith row of $\mathbf{X}$.

Theorem 5 (Consistency). *Assume conditions (C1)–(C2) hold. Suppose $\lambda_n \to 0$, $n\lambda_n \to \infty$, and $d_n \to d_0 \in (0,1)$. Then*

$$\hat{\boldsymbol{\beta}}^{\textit{Liu-SCAD}}(\lambda_n, d_n) \xrightarrow{p} \boldsymbol{\beta}_0.$$

We aim to prove that the Liu-SCAD estimator

$$\hat{\boldsymbol{\beta}}^{\text{Liu-SCAD}}(\lambda_n, d_n) = \arg\min_{\boldsymbol{\beta}\in\mathbb{R}^p} Q_n(\boldsymbol{\beta}),$$

with

$$Q_n(\boldsymbol{\beta}) := \frac{1}{2n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \frac{1}{2}\|\boldsymbol{\beta} - d_n\hat{\boldsymbol{\beta}}^{\text{LS}}\|_2^2 + \sum_{j=1}^{p} p_{\lambda_n}^{\text{SCAD}}(|\beta_j|),$$

converges in probability to the unique minimizer of the limiting function

$$L_0(\boldsymbol{\beta}) := \frac{1}{2}(\boldsymbol{\beta} - \boldsymbol{\beta}_0)^\top\mathbf{C}(\boldsymbol{\beta} - \boldsymbol{\beta}_0) + \frac{1}{2}\|\boldsymbol{\beta} - d_0\boldsymbol{\beta}_0\|_2^2,$$

as $n \to \infty$, under the assumptions (C1)–(C2), $\lambda_n \to 0$, $n\lambda_n \to \infty$, and $d_n \to d_0 \in (0,1)$.

We first analyze the behavior of each term in $Q_n(\boldsymbol{\beta})$. Define

$$R_n(\boldsymbol{\beta}) := \frac{1}{2n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2.$$

Using the model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon}$, we have

$$R_n(\boldsymbol{\beta}) = \frac{1}{2n}\|\mathbf{X}(\boldsymbol{\beta} - \boldsymbol{\beta}_0) - \boldsymbol{\varepsilon}\|^2.$$

Expanding the square, we obtain

$$R_n(\boldsymbol{\beta}) = \frac{1}{2n}(\boldsymbol{\beta} - \boldsymbol{\beta}_0)^\top \mathbf{X}^\top \mathbf{X}(\boldsymbol{\beta} - \boldsymbol{\beta}_0) - \frac{1}{n}\boldsymbol{\varepsilon}^\top \mathbf{X}(\boldsymbol{\beta} - \boldsymbol{\beta}_0) + \frac{1}{2n}\|\boldsymbol{\varepsilon}\|^2.$$

Under assumption (C1), we know that

$$\frac{1}{n}\mathbf{X}^\top \mathbf{X} \to \mathbf{C},$$

so the first term converges uniformly on compact subsets to

$$\frac{1}{2}(\boldsymbol{\beta} - \boldsymbol{\beta}_0)^\top \mathbf{C}(\boldsymbol{\beta} - \boldsymbol{\beta}_0).$$

For the second term, by the law of large numbers, the cross term satisfies

$$\sup_{\|\boldsymbol{\beta} - \boldsymbol{\beta}_0\| \le M} \left| \frac{1}{n}\boldsymbol{\varepsilon}^\top \mathbf{X}(\boldsymbol{\beta} - \boldsymbol{\beta}_0) \right| = o_p(1),$$

since the random vector $\boldsymbol{\varepsilon}$ has zero mean, finite variance, and under (C2), the row norms of $\mathbf{X}$ are asymptotically negligible.

The third term $\frac{1}{2n}\|\boldsymbol{\varepsilon}\|^2$ does not depend on $\boldsymbol{\beta}$, so it does not affect the minimization and can be ignored in the limit.

Now we consider the Liu shrinkage term. Since $\hat{\boldsymbol{\beta}}^{\mathrm{LS}} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{y} \xrightarrow{p} \boldsymbol{\beta}_0$, and $d_n \to d_0$, we conclude that

$$\|\boldsymbol{\beta} - d_n \hat{\boldsymbol{\beta}}^{\mathrm{LS}}\|^2 \xrightarrow{p} \|\boldsymbol{\beta} - d_0 \boldsymbol{\beta}_0\|^2,$$

uniformly on compact sets.

Next, consider the SCAD penalty. Since $\lambda_n \to 0$ and the SCAD penalty satisfies $p_{\lambda_n}^{\mathrm{SCAD}}(|\beta_j|) \to 0$ for any fixed β_j, we have

$$\sup_{\boldsymbol{\beta} \in K} \sum_{j=1}^{p} p_{\lambda_n}^{\mathrm{SCAD}}(|\beta_j|) \to 0$$

for every compact $K \subset \mathbb{R}^p$.

Putting all parts together, we conclude that

$$\sup_{\boldsymbol{\beta}\in K} |Q_n(\boldsymbol{\beta}) - L_0(\boldsymbol{\beta})| \xrightarrow{p} 0,$$

for every compact set $K \subset \mathbb{R}^p$, which means $Q_n \xrightarrow{p} L_0$ uniformly on compacts.

To complete the argument, we note that $L_0(\boldsymbol{\beta})$ is strictly convex due to the positive definiteness of $\mathbf{C}$, so it admits a unique global minimizer in $\mathbb{R}^p$.

By an application of the argmin consistency theorem (e.g., Theorem 5.7 of van der Vaart [20]), we conclude that

$$\hat{\boldsymbol{\beta}}^{\text{Liu-SCAD}}(\lambda_n, d_n) \xrightarrow{p} \arg\min_{\boldsymbol{\beta}} L_0(\boldsymbol{\beta}).$$

This result confirms that the Liu-SCAD estimator is consistent under mild conditions, blending the oracle-like sparsity recovery of SCAD with the regularization stability of the Liu-type shrinkage framework.

B.2 Oracle Properties of the Liu-SCAD Estimator

Let $\boldsymbol{\beta}_0 = (\boldsymbol{\beta}_{01}^\top, \boldsymbol{\beta}_{02}^\top)^\top$, where $\boldsymbol{\beta}_{01} \in \mathbb{R}^{q_n}$ contains the nonzero elements and $\boldsymbol{\beta}_{02} = \mathbf{0} \in \mathbb{R}^{p_n - q_n}$. Assume $q_n = o(n)$ and $p_n < n$. Define $\mathbf{X}_1$ as the subdesign matrix corresponding to $\boldsymbol{\beta}_{01}$ and $\Sigma_{n1} = \frac{1}{n}\mathbf{X}_1^\top \mathbf{X}_1$.

We impose the following conditions:

(A1) ε_i are i.i.d. with $\mathbb{E}[\varepsilon_i] = 0$ and $\mathbb{E}[\varepsilon_i^2] = \sigma^2 < \infty$.

(A2) The eigenvalues of $\frac{1}{n}\mathbf{X}^\top \mathbf{X}$ are bounded away from zero and infinity uniformly in n.

(A3) The minimum signal strength $b_{0n} = \min_{1\le j\le q_n} |\beta_{0j}|$ satisfies $(p_n/n)^{1/2}/b_{0n} \to 0$. Also, $\lambda_n \to 0$, $n\lambda_n \to \infty$, and $d \in (0,1)$ is fixed.

Theorem 6 (Oracle Property). *Under (A1)–(A3), the Liu-SCAD estimator $\hat{\boldsymbol{\beta}}^{\text{Liu-SCAD}}$ satisfies:*

(i) (Sparsity) $\Pr(\hat{\boldsymbol{\beta}}_2 = 0) \to 1$.

(ii) (Asymptotic normality) For any $\mathbf{a}_n \in \mathbb{R}^{q_n}$ with $\|\mathbf{a}_n\|_2 = 1$,

$$\sqrt{n} \cdot \mathbf{a}_n^\top (\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_{01}) \xrightarrow{d} \mathcal{N}(0, \sigma^2 \cdot \mathbf{a}_n^\top \Sigma_{n1}^{-1} \mathbf{a}_n).$$

Let $\boldsymbol{\beta}_0 = (\boldsymbol{\beta}_{01}^\top, \boldsymbol{\beta}_{02}^\top)^\top$ be the true parameter vector, with $\boldsymbol{\beta}_{02} = \mathbf{0}$ representing the indices of irrelevant covariates. Let $\hat{\boldsymbol{\beta}}^{\text{Liu-SCAD}} = (\hat{\boldsymbol{\beta}}_1^\top, \hat{\boldsymbol{\beta}}_2^\top)^\top$ denote the Liu-SCAD estimator obtained as the minimizer of the following objective function:

$$Q_n(\boldsymbol{\beta}) = \frac{1}{2n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \frac{1}{2}\|\boldsymbol{\beta} - d\hat{\boldsymbol{\beta}}^{\text{LS}}\|_2^2 + \sum_{j=1}^{p_n} p_{\lambda_n}^{\text{SCAD}}(|\beta_j|),$$

where $d \in (0,1)$ is fixed and $\hat{\boldsymbol{\beta}}^{\text{LS}}$ is the ordinary least squares estimator.

We prove parts (i) and (ii) of the theorem sequentially.

(i) Sparsity: We show that with probability tending to one, $\hat{\boldsymbol{\beta}}_2 = 0$.

Since $\boldsymbol{\beta}_{02} = \mathbf{0}$ and $b_{0n}^{-1}(p_n/n)^{1/2} \to 0$ by assumption (A3), the irrelevant components are sufficiently small compared to the penalty level. Moreover, the SCAD penalty satisfies the unbiasedness property: for small β_j, the derivative $p'_{\lambda_n}(|\beta_j|)$ behaves like λ_n.

Consider the Karush–Kuhn–Tucker (KKT) conditions for minimization. For j corresponding to a zero component of $\boldsymbol{\beta}_0$, the subgradient condition is

$$\left|\frac{1}{n}\mathbf{x}_j^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}^{\text{Liu-SCAD}}) + (\hat{\beta}_j - d\hat{\beta}_j^{\text{LS}})\right| \le p'_{\lambda_n}(0+).$$

Since $\hat{\beta}_j^{\text{LS}} = O_p(1)$ and $d \in (0,1)$, the shrinkage term is $O_p(1)$. Under assumption (A1) and the scaling of λ_n, the stochastic part $\frac{1}{n}\mathbf{x}_j^\top \boldsymbol{\varepsilon}$ is $O_p(n^{-1/2})$. Given that $n\lambda_n \to \infty$, the right-hand side $p'_{\lambda_n}(0+) = \lambda_n$ dominates this term asymptotically, implying that the inequality holds with high probability, and thus $\hat{\beta}_j = 0$.

Therefore, with probability tending to one, all coefficients corresponding to zero true coefficients are exactly zero in $\hat{\boldsymbol{\beta}}^{\text{Liu-SCAD}}$. That is,

$$\mathbb{P}(\hat{\boldsymbol{\beta}}_2 = 0) \to 1.$$

(ii) Asymptotic Normality: Now we prove that the nonzero components of $\hat{\boldsymbol{\beta}}^{\text{Liu-SCAD}}$ are asymptotically normal.

Condition on the event $\hat{\boldsymbol{\beta}}_2 = 0$, which holds with probability tending to one. On this event, the estimator reduces to minimizing the following restricted objective over $\mathbb{R}^{q_n}$:

$$Q_n^{(1)}(\boldsymbol{\beta}_1) = \frac{1}{2n}\|\mathbf{y} - \mathbf{X}_1\boldsymbol{\beta}_1\|_2^2 + \frac{1}{2}\|\boldsymbol{\beta}_1 - d\hat{\boldsymbol{\beta}}_1^{\text{LS}}\|_2^2 + \sum_{j=1}^{q_n} p_{\lambda_n}^{\text{SCAD}}(|\beta_j|).$$

Since $\lambda_n \to 0$ and $b_{0n}^{-1}(p_n/n)^{1/2} \to 0$, the SCAD penalty is asymptotically inactive on the nonzero components (due to the flatness of the SCAD function for large $|\beta_j|$). Therefore, the penalized estimator behaves like an unpenalized estimator with shrinkage toward $d\hat{\boldsymbol{\beta}}_1^{\text{LS}}$.

Let us denote:

$$\hat{\boldsymbol{\beta}}_1^{\text{Liu-SCAD}} = \arg\min_{\boldsymbol{\beta}_1} \left\{ \frac{1}{2n}\|\mathbf{y} - \mathbf{X}_1\boldsymbol{\beta}_1\|_2^2 + \frac{1}{2}\|\boldsymbol{\beta}_1 - d\hat{\boldsymbol{\beta}}_1^{\text{LS}}\|_2^2 \right\}.$$

The first-order condition gives:

$$\left(\frac{1}{n}\mathbf{X}_1^\top\mathbf{X}_1 + I\right)\hat{\boldsymbol{\beta}}_1^{\text{Liu-SCAD}} = \frac{1}{n}\mathbf{X}_1^\top\mathbf{y} + d\hat{\boldsymbol{\beta}}_1^{\text{LS}}.$$

Substituting $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_{01} + \boldsymbol{\varepsilon}$ and using standard expansions, we get:

$$\hat{\boldsymbol{\beta}}_1^{\text{Liu-SCAD}} - \boldsymbol{\beta}_{01} = (\Sigma_{n1} + I)^{-1}\left(\frac{1}{n}\mathbf{X}_1^\top\boldsymbol{\varepsilon} + (d-1)(\hat{\boldsymbol{\beta}}_1^{\text{LS}} - \boldsymbol{\beta}_{01})\right).$$

Using the approximation $\hat{\boldsymbol{\beta}}_1^{\text{LS}} - \boldsymbol{\beta}_{01} = (\mathbf{X}_1^\top\mathbf{X}_1)^{-1}\mathbf{X}_1^\top\boldsymbol{\varepsilon}$, the dominant term becomes:

$$\hat{\boldsymbol{\beta}}_1^{\text{Liu-SCAD}} - \boldsymbol{\beta}_{01} = (\Sigma_{n1} + I)^{-1}\left[\left(\frac{1}{n}\mathbf{X}_1^\top\boldsymbol{\varepsilon}\right) + (d-1)\left(\Sigma_{n1}^{-1}\frac{1}{n}\mathbf{X}_1^\top\boldsymbol{\varepsilon}\right)\right] + o_p(n^{-1/2}).$$

Collecting terms and applying Slutsky's theorem and the Lindeberg-Feller CLT (under A1), for any sequence $\mathbf{a}_n$ with $\|\mathbf{a}_n\|_2 = 1$,

$$\sqrt{n}\cdot\mathbf{a}_n^\top(\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_{01}) \xrightarrow{d} \mathcal{N}\left(0, \sigma^2\cdot\mathbf{a}_n^\top\Sigma_{n1}^{-1}\mathbf{a}_n\right),$$

since the preconditioning matrix converges to the identity as $n \to \infty$ under (A2). This completes the proof of both sparsity and asymptotic normality.

B.3 Grouping Property of the Liu-SCAD Estimator

Theorem 7 (Grouping Property). *Assume the columns of* $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_p)$ *are standardized so that* $\sum_{i=1}^n x_{ij} = 0$ *and* $\|\mathbf{x}_j\|_2^2 = n$ *for all* j*. Then for any* $i \neq j$*, if* $\frac{1}{n}\mathbf{x}_i^\top\mathbf{x}_j = \rho_{ij}$ *and* $\hat{\beta}_i, \hat{\beta}_j \neq 0$*, the Liu-SCAD estimator satisfies*

$$|\hat{\beta}_i - \hat{\beta}_j| \leq C\cdot\sqrt{2(1-\rho_{ij})},$$

where $C > 0$ *depends on* λ*,* d*,* $\|\mathbf{y}\|_2$*, and the SCAD parameters.*

The bound follows from the shrinkage effect of the Liu term, which encourages similar coefficients when predictors are highly correlated. The derivation uses coordinate-wise updates, Taylor expansion, and the Cauchy–Schwarz inequality.

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